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CFD applications in various heat exchangers design: A review

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ABSTRACT

This literature review focuses on the applications of Computational Fluid Dynamics (CFD) in the field of heat exchangers. It has been found that CFD has been employed for the following areas of study in various types of heat exchangers: fluid flow maldistribution, fouling, pressure drop and thermal analysis in the design and optimization phase. Different turbulence models available in general purpose commercial CFD tools i.e. standard, realizable and RNG $k-\varepsilon$ RSM, and SST $k-\varepsilon$ in conjunction with velocity-pressure coupling schemes such as SIMPLE, SIMPLEC, PISO and etc. have been adopted to carry out the simulations. The quality of the solutions obtained from these simulations are largely within the acceptable range proving that CFD is an effective tool for predicting the behavior and performance of a wide variety of heat exchangers.

1. Introduction

Heat exchangers play a significant role in the operation of many systems such as power plants, process industries and heat recovery units. Its inevitable need has necessitated work on efficient and reliable designs leading towards optimum share in the overall system performance. The Log Mean Temperature Difference (LMTD) method and the number of heat transfer units (NTU) method have been used for heat exchanger design [1]. These methods have some shortcomings associated with them i.e. iterative in nature and need of a prototype to implement the design. Due to these reasons, these methods are time consuming as well as expensive especially for large scale models. However, economical access to powerful micro processors has paved the way for evolvement of Computational Fluid Dynamics (CFD) during the design phase.

CFD is a science that can be helpful for studying fluid flow, heat transfer, chemical reactions etc by solving mathematical equations with the help of numerical analysis. It is equally helpful in designing a heat exchanger system from scratch as well as in troubleshooting/optimization by suggesting design modifications. CFD employs a very simple principle of resolving the entire system in small cells or grids and applying governing equations on these discrete elements to find numerical solutions regarding pressure distribution, temperature gradients, flow parameters and the like in a shorter time at a lower cost because of reduced required experimental work [2,3].

The results obtained with the CFD are of acceptable quality [4–9]. In the current work, various problems encountered in the design of heat exchangers and their solutions with the help of CFD have been reviewed. This work can serve as a ready reference for application of CFD in design of various types of heat exchangers.

2. Types of heat exchangers

There are innumerable types of heat exchangers in use including but not limited to plate type heat exchanger (PHE), shell-and-tube heat exchanger, vertical mantle heat exchanger and micro heat exchanger and a broad classification of heat exchangers based on their construction is outlined in the Fig. 1 below:

3. Applications of CFD in various aspects of heat exchangers

Some of the commercial CFD codes in use are FLUENT, CFX, STAR CD, FIDAP, ADINA, CFD2000, PHOENICS and others [10]. Using these, extensive work has been carried out on the various aspects of heat exchangers with primary focus on optimization through study

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Nomenclature

BSL_RSM	Baseline Reynolds stress model
c	Thermal capacity (J/kg K)
d_e	Hydraulic diameter (m)
De	Dean Number
f	Fanning friction factor (–)
f'	Friction factor (–)
G	Mass flow rate (kg/(m^2 s))
GAMBIT	Genomic Analysis and Mapping by In Vitro Transposition
h	Heat transfer coefficient (W/m^2 K)
j	Colburn j-factor (–)
k	Thermal conductivity (W/m K)
L	Length (m)
LMTD	Log mean temperature difference
NTU	Number of heat transfer units
Nu	Nusselt Number
ΔP	Pressure drop
P	Pressure, N/m^2
PCHE	Printed circuit board heat exchanger
PISO	Pressure-Implicit with Splitting of Operators
$-p$	In a pitch
Pr	Prandtl Number
Q	Heat transferred (W)
q	Heat source (W/m^3)
r	Radial coordinate (m)
R	Resistance to the flow of thermal energy (m^2 K/W)
Ra	Rayleigh Number
Re	Reynolds Number
RNG	Renormalization group
RSM	Reynolds stress model
SIMPLE	Semi Implicit Method for Pressure Linked Equations
SIMPLEC	Semi Implicit Method for Pressure Linked Equations-Consistent
T	Temperature (K)
t	Time (s)
U	Overall heat transfer coefficient (W/m^2 K^1)
u	Velocity (m/s)
v	Fluid flow velocity (m/s)
V	Volume (m^3)
X	Sensitivity coefficient (–)
x, y, z	Spatial coordinates (m)
$\varnothing_1$	Generalized transport variable for header 1
$\varnothing_2$	Generalized transport variable for header 2
ρ	Density (kg/m^3)

of thermal performance, pressure drop, fouling and fluid maldistribution [1,5,11–14]. Bengt Sunde'n [15] has listed various possible models that might be utilized in the CFD analysis and carried out the simulations on a PHE with different corrugations to show the possibility of application of CFD on various problems encountered in heat exchangers.

In the subsequent sections, various issues related to heat exchangers have been presented under different headings. Under each heading, exchangers have been discussed in order of their types.

3.1. Flow maldistribution

Non-uniformity in fluid flow is one of the primary reasons resulting in a poor heat exchanger performance. This may be attributed to improper design of inlet/outlet port and header configuration, distributor construction and plate corrugations [16]. A number of researchers as shown in Table 1 below have approached the maldistribution problem with CFD by employing various turbulence models and integrating schemes focusing on pressure variation and velocity distribution.

To understand the fluid flow maldistribution, L. J. Shah et al. [17] studied the characteristics of vertical mantle heat exchanger employed in a solar water heater. A three block model was considered in the analysis which revealed that fluid distribution along the mantle is affected by recirculation produced due to buoyancy force in the mantle in case of high and low temperature inputs respectively. A local Nusselt–Rayleigh correlation was worked out, for which x is the distance from top of annulus:

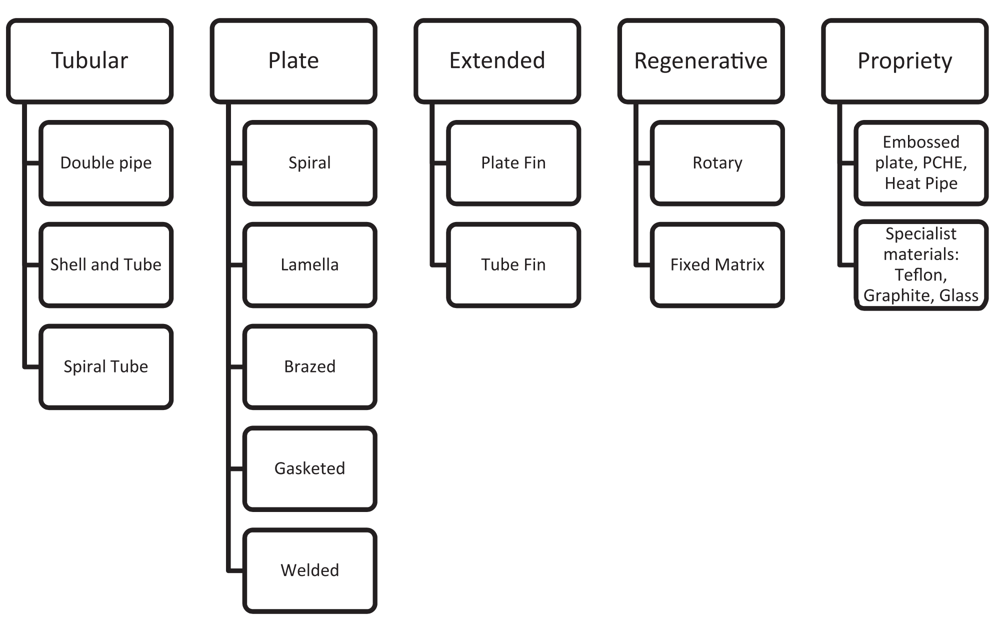

Fig. 1. Classification of Heat Exchangers on the basis of construction.

Table 1
CFD simulations on heat exchangers in fluid flow maldistribution.

Authors/Type	CFD Methodology	Comparison between Experimental and Simulation Results
L. J. Shah et al. (2000) Vertical Mantle Heat Exchanger	• CFD code: CFX • Pressure-Velocity Coupling Scheme: SIMPLE	Close agreement observed
Koen Grijspeerdt et al. (2003) Plate Heat Exchanger	• CFD Code: FINE-Turbo, EURANUS code • Turbulence Model: Baldwin–Lomax turbulent stress model • Meshing software: FINE-Turbo pre-processor	–
Thomas Perrotin et al. (2004) Louvered Fin and Flat Tube Heat Exchanger	• Turbulence Model: $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLE, PISO • Mesh: Quadrilateral	13%
Zhe Zhang et al.(2003) Plate Fin Heat Exchanger	• CFD Code: FLUENT • Pressure-Velocity Coupling Scheme: SIMPLE • Turbulence Model: Standard $k-\varepsilon$ • Mesh: finest mesh of 150,000 cells	Good agreement observed
Jian et al. (2004) Plate-Fin Heat Exchanger	• CFD Code: FLUENT • Pressure-Velocity Coupling Scheme: Semi Implicit SIMPLER, Second Order Upwind • Turbulence Model: Two equation $k-\varepsilon$ • Mesh: finest mesh of 245,817 cells	Close agreement observed
S. Knudsen et al. (2004) Vertical Mantle Heat Exchanger	• CFD Code: FLUENT 6.0 • Pressure-Velocity Coupling Scheme: PISO and Second Order Upwind • Mesh: finest mesh of 193,150 cells	Good agreement observed
A. G. Kanaris et al. (2005) Narrow Channels with Corrugated Walls	• CFD Code: CFX • Turbulence Models: Two equation SST • Pressure-Velocity Coupling Scheme: SIMPLEC, QUICK • Meshing Software: CFX 5.6 and ICEM CFD 4.CFX • Mesh: unstructured tetrahedral	Good conformity with literature results
Vimal Kumar et al. (2005) Tube-in-Tube Helical Heat Exchanger	• CFD Code: FLUENT 6.0 • Turbulence Model: $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLEC	No such comparison available
Carla S. Fernandes et al. (2007) Double Sin Chevron Plate Heat Exchanger	• CFD Code: POLYFLOW • Pressure-Velocity Coupling Scheme: Semi Implicit SIMPLER • Turbulence Model: FVM and two equation $k-\varepsilon$ • Meshing Software: GAMBIT • Mesh: Unstructured mesh of tetrahedral, hexahedral and pyramidal cells	Less than 4%
Kilas et al. (2007) Plate Heat Exchanger	• CFD Code: FLUENT 6.1 • Segregated solver • Turbulence Model: $k-\varepsilon$ • Meshing software: GAMBIT v2.1 • Mesh: Structured tri-tetra grid of 2,000,000 cells	Close agreement observed
Mourad Yataghene et al.(2007) Scraped Surface Heat Exchanger	• CFD Code: FLUENT • Meshing Software: GAMBIT • Mesh: Hybrid	Close agreement observed
Myoung Il Kim et al. (2008) Shell and Tube Type Heat Exchanger	• CFD Code: FLUENT 6.0 • Turbulence Model: $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLEC • Mesh: Tetrahedral	A good agreement observed
Li-Zhi Zhang et al. (2009) Cross Flow Air to Air Exchanger	• CFD Code: FLUENT • Pressure-Velocity Coupling Scheme: SIMPLE • Meshing software: GAMBIT • Mesh: Coarse (tetrahedral and hexahedral 160,837 cells)	–
C. T'Joen et al. (2010) Inclined Louvered Fin Type Exchanger	• CFD Code: FLUENT	A good agreement observed
David A. Yashar et al. (2011) Louvered Fin Tube Exchanger	• Turbulence Model: Momentum Resistance Approach using $k-\varepsilon$ model	Generally less than 3% Up to 10% for large rise in velocity

$$\mathrm{Nu}_x = 0.46 \times \left(\mathrm{Ra}_x \times \frac{x}{D_{\mathrm{hydraulic}}}\right)^{0.28} \quad \text{for Ra} = \mathrm{Ra}_x \le 10^{11} \qquad (1)$$

In a similar exchanger, S. Knudsen et al. [18] observed presence of mixed flow in the mantle close to inlet by studying the flow pattern inside tank and mantle of the exchanger.

In 2002, Koen Grijspeerdt et al. [16] established that 2D analysis is sufficient to identify the effect of the corrugations shape, but to investigate on the significance of the corrugation orientation, a 3D analysis is also mandatory. For the purpose, a numerical model was constructed and simulation performed between two corrugated plates of a PHE keeping Re and Pr 4482 and 6.62 respectively. It was observed that recirculation bubbles are formed, which acts as a hindrance to the steady flow. Carla S. Fernandes et al. [19] studied a double sine-chevron PHE for a range of corrugation angles (i.e. 30° to 85°) and channel aspect ratios (i.e. 0.38–0.76). The shape factor and the Tortuosity coefficient were found to depend on the corrugation angle and channel aspect ratios respectively. An increase in Tortuosity coefficient and Kozeny's coefficient was observed with the increase in the channel aspect ratio which eventually decreased with the increase in chevron plate angle. It was deduced that shape factor is almost independent of channel aspect ratios and inversely proportional to chevron plate angles.

Louvered fin and flat tube heat exchangers have widespread application in automotive and air conditioning industry. Thomas Perrotin et al. [20] carried a study for 2D and 3D simulation. The authors observed, for lower Re (i.e. 78.6), air makes a thin boundary layer with the louver due to which the space between the successive louvers is blocked. Because of the hampered air flow path, the temperature of the air in that local region increases which reduces the overall heat transfer rate. On the contrary, at higher Re, the boundary layer is found to be thin and the air velocity is aligned with the louvers. C. T'Joen et al. [21] investigated the relation between the mean flow of the fluid and the thermal-hydraulic behavior using dye injection technique for different configurations and a range of Re. The flow parameters were analyzed using fin angle-alignment factor; a decrease in the fin angle produces a more uniform flow resulting in an increase in f and j. Recirculation zones were observed at the inclined section of the fin for low Re and study concluded that inclined louvered fins give better results than slit or louvered fins. David A. Yashar et al. [22] considered a vertical, single slab, four depth row exchanger to compare the air flow distribution measured through CFD and particle image velocimetry. The flow distribution was modeled in terms of pressure variations, fluctuating at the bracket mount, forming regions with velocities ranging from 1.25 ms^{-1} to 1.7 ms^{-1}. The agreement between the CFD and the experimental results presented momentum resistance modeling methodology as a viable means to analyze the flow in finned tube exchangers.

Zhe Zhang et al. [23], 2003, simulated the inlet configuration to optimize the header design in a plate fin heat exchanger. The ratio of equivalent diameter was used, which is defined as:

$$\beta = \frac{\left[\phi_{1,\mathrm{out}}/\phi_{2,\mathrm{in}}\right]}{\left[\phi_{2,\mathrm{out}}/\phi_{1,\mathrm{in}}\right]} \tag{2}$$

Here ϕ is a generalized transport variable. The results obtained with the help of CFD showed that the flow distribution is critical in the y direction, shown in Fig. 2, for the header design currently in use in the industry. Based on this conclusion, two optimized headers having two-stage-distributing structures were proposed and verified through simulation. It was established that a more uniform flow may be obtained if the ratios are kept equal for the outlet and inlet equivalent diameters. The very next year, Jian et al. [24] proposed to install a baffle with holes to increase the uniformity in fluid flow and consequently, the effectiveness of the PHE. The results of the simulations indicated that employing baffles with smaller holes decreases the maldistribution factor from 0.95 to

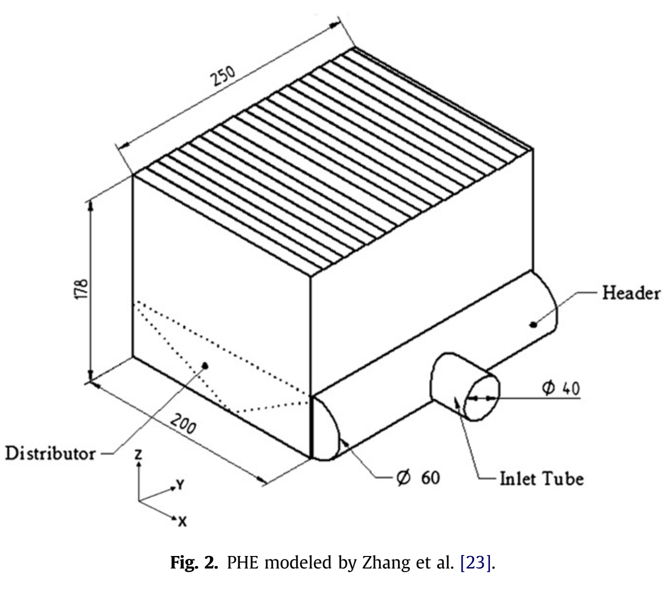

Fig. 2. PHE modeled by Zhang et al. [23].

0.32. Kilas et al. [11] concluded a decrease in the flow non-uniformity from 15.5 to 4.5, i.e. 70.71% reduction, using the segregated solver approach on a tri-tetra grid to analyze the performance and velocity distribution at the inlet section. A non-uniform velocity pattern was observed in the y-direction having a maximum value at the center and decreasing along the side ways. As a remedy, a larger header height was incorporated.

Mourad Yataghene et al. [25] considered a scraped surface heat exchanger for Newtonian and non Newtonian fluids using electrochemical procedure and validated the experimental results with the 2D numerical simulation with a hybrid mesh along with refined grids in between the tips. Consideration was given to the lean gaps meshing and it was found that shear rate is maximum between tip of the blade and the stator. Considering the results for fluids namely HV45, CMC and Guar Gum, the author stated that the shear rate mainly depends on the gap between the blade tip and the outer drum; it increases if the gap is reduced. It was established that a 3D simulation is essential to analyze the temperature distribution completely.

In the case of corrugated narrow channel heat exchanger, A. G. Kanaris et al. [26] reported the significant role played by the side channels in the fluid flow. It was observed that fluid flows mainly towards the right side channel and after striking the side wall, it moves on to the opposite channel while some also passes over the top crest of the channels. A similar type of flow behavior was observed by Focke et al. [27]. Two corrugated plates were considered without the side channels and it was noticed that after hitting the side wall, fluid moves backward and vortices are formed with the in-coming fluid.

Counter flow in a tube-in-tube helical heat exchanger was simulated by Vimal Kumar et al. [1]. The flow through both the tubes was studied for various known cross-sections through the control volume finite difference method. It was found that for small axial distance, effect of secondary flow due to centrifugal force is minimum whereas with an increased axial distance, the centrifugal force pushes the fluid outward making secondary flow maximum across the walls of the inner tube. Same result was observed for the outer tube where, for small axial distance, the effect of secondary flow is weak but for larger axial distance it reaches the upper limit at the outer wall.

Myoung Il Kim et al. [28] estimated the flow pattern characteristics of the shell-and-tube type heat exchanger. Different header types were considered and results indicated that A-type having the smallest header length and minimum flow rate (655 mm and 0.54 $m^3 s^{-1}$) is not sufficient in distributing the flow uniformly along the length whereas C-type design (1092.5 mm and 1.62 $m^3 s^{-1}$) yields the best results. Moreover, flow becomes more uniform when the length of the header is increased and correspondingly decreases with reduction in the flow rate of working media. The shape and configuration of the inlet nozzle also affects the flow pattern of header. Ender Ozden et al. [10], for the same heat exchanger stated that few recirculation regions are formed at the rear of the baffles when the number of baffles is small. Increased baffle numbers can improve the rate of heat transfer.

Next year, Li-Zhi Zhang [12] analyzed a cross flow air to air heat exchanger with plate-fin cores. Prior to them, Miura et al. [29], Galeazzo et al. [30] and Bansode et al. [31] studied flow maldistribution using different number of flow channels. Keeping the previous results in view, Zhang assumed three channel pitches: 1.8 mm (B), 2.5 mm (A), and 4 mm (C) for a total air flow stream of 150 m^3/hr. The maximum degree of mal flow distribution occurred in the channel with the largest core pitch [12]. In case of a staggered elliptical tube heat exchanger, K. Kirtikos et al. [4] observed that the type of fluid flow in the last set of tubes has a very minute effect on the heat transfer property.

3.2. Fouling

Degradation of the working fluid because of high temperatures or interaction with the exchanger walls results in production of by products which get deposited on the walls and reduce the coefficient of heat transfer. Frequent periodic maintenance may be required if extreme cases of fouling are encountered. This phenomenon is frequently encountered in the food processing industry largely involving PHE's and have been investigated by using CFD by various authors as shown in Table 2.

Koen Grijspeerdt et al. [16], 2002, through simulation for temperature profiles of milk inside a PHE pointed out weak regions formed due to fouling by Beta-lectoglobulin, which primarily relies on the difference of temperature between the plate and the working media. Soojin Jun et al. [5] investigated the impact of corrugations and their position on fouling rates for a similar PHE used for the pasteurization of milk. An alternate design of a compact heat exchanger was proposed in an effort to establish the possibility of using unconventional heat exchangers in the food processing industry. It was found that area requirements are decreased for the same amount of heat transfer along with a reduction in deposition rate by 92.6%.

In addition to reinstating the fact fouling rate depends on wall temperatures, Maria Valeria et al. [32], 2009, studied the fouling rates with respect to flow velocities. As the Re of flowing media is increased, a drastic increase in the fouling rate is encountered with a deposition of a little more than 1 g for Re 1700 and almost 22 g for Re 3700 after a given time period. Thus, through the research, it was established that basic computation may be used for a variety of geometry and biochemical products.

3.3. Pressure drop

Every fluid undergoes some pressure loss while flowing in the heat exchanger. This phenomenon is mainly affected by the choice of design of core or the matrix and the fluid distribution components i.e. inlet/outlet ports, headers, nozzles, ducts and etc [33]. This pressure loss particularly because of flow distribution devices seriously affects the heat transfer rate of the exchanger. CFD has facilitated the investigation of this phenomenon as shown in Table 3 below.

G. Gan et al. [34] performed a CFD simulation on the tubes of a heat exchanger used in closed-wet cooling towers. Pressure drop was found to depend on the tube configurations and water to air ratio. Predicted pressure loss coefficient was found inversely proportional to transverse pitch but was is in direct relationship with water to air mass flow rate. Moreover, pressure loss was seen to increase by 17% with the interference of upstream air, but a reduced value was obtained when this air is bypassed. Vikas Kumar et al. [35] observed the pressure to be maximum at the separation plate of hot and cold streams in the vertical plane of a cross flow air to air heat exchanger. The cold air pressure in the tube was found to be highest at the inlet that decreased gradually by 202 Pa due to friction with the tube inner wall.

In 2007, Q.W Dong et al. [36] considered a rod baffle shell and tube heat exchanger. A periodic flow unit duct model was used for examining the pressure drop. Simulations were carried out using water as the working fluid at 300 K while for the experimental setup, laser Doppler anemometry was opted to compute flow velocities. Pressure drop was found to have an inverse relationship with the baffle pitch assuming flow velocity to be constant whereas it was seen to increase with the rise in fluid velocity at constant baffle pitch. The results concluded that the periodic duct unit model can define the flow and pressure characteristics more precisely than other models. Similarly, Yong Qing Wang et al. [3] established that amongst small round hole plate, plum blossom plate, rectangle hole plate, big round hole plate, rod baffle and ting baffle, rod baffle is the most reliable and suitable supporting structure with a minimum pressure drop. Qiuwang Wang et al. [37] stated the pressure drop of a combined multiple shell-pass shell-and-tube heat exchanger (CMSP-STHX) to be lower than shell-and-tube heat exchanger (SG-STHX) with segmental baffles, for the same overall heat transfer, by almost 13%. Mohammadi et al. [38] studied the effect of baffle cut and location on an exchanger having 660 tubes while neglecting the effect of leakage flows. Carrying the study forward, in 2009, Mohammadi et al. [39] undertook another simulation with the leakage flows accounted for as well. The employed exchanger was in accordance with the TEMA standards with 76 plane tubes. Two different baffle orientations were considered and three different fluids i.e. air, water and engine oil used to evaluate the effect of viscosity for a Pr in the range of 0.7–206. The CFD results established that a horizontal baffle exchanger has a 20% higher heat transfer coefficient with 250% greater pressure drop than the one with vertical baffles.

Tri Lam Ngo et al. [40] simulated the pressure drop in an S-shaped finned micro channel heat exchanger which was found 6–7 times lower than the ordinary zigzagged finned exchanger. An experimental study was performed to investigate the performance (pressure drop and Nu characteristic) of the two different fins respectively. A four to five times lesser pressure drop was observed experimentally in the micro channel heat exchanger while keeping the Nu 40% lower. Pr was established to be independent in the range 0.75–2.2 from Nu and pressure drop factor, provided Re varied between $3.0 \times 103–2.0 \times 104$.

Considering the results of Ishizuka et al. [41,42], in 2008, Dong Eok Kim et al. [43] proposed to replace the zigzag channels with airfoil type finned channels for a printed circuit board heat exchanger carrying supercritical carbon dioxide. The new design reduced the pressure drop while giving same rate of heat transfer as the conventional design.

Yin Chi Tasai et al. [44] stated that the pressure drop in two cross corrugated channels of a chevron PHE increases by 10,000 Pa via simulation analysis while a figure of 12,500 Pa is obtained for experimental study as the Re of the fluid is increased from 600 to 1650. Similarly, Xiao-Hong Han et al. [45] observed a decrease in the pressure along the direction of the flow for a chevron PHE with fluid velocity being very low at the corrugated side of the inflow and outflow port. On the other hand, Yong Qing Wang et al. [6] established that the pressure drop and heat transfer coefficient for the plate fin heat exchanger with plain fins is smaller than serrated fins for the same Re. For the purpose, they simulated a simplified model of one fin layer of the exchanger using a segregated solver approach and SIMPLE algorithm. The model was

Table 2
CFD simulations on heat exchangers in fouling.

Authors/Type	CFD Methodology	Comparison between Experimental and Simulation Results
Soojin Jun et al. (2005) Plate Heat Exchanger	• CFD Code: FLUENT 6.0 (Arrhenius rate expressions) • Meshing Software: GAMBIT • Mesh: Tetrahedral (831169 cells)	Within 2%
Maria Valeria et al. (2009) Plate Heat Exchanger	• Pressure-Velocity Coupling Scheme: SIMPLEX • Environment mode: FRONTIER	Found to be in good agreement

Table 3
CFD simulations on heat exchangers in pressure drop.

Authors/Type	CFD Methodology	Comparison between Experimental and Simulation Results
G. Gan et al. (2000) Heat Exchanger For Closed Wet Cooling Towers	• CFD Code: FLUENT 4.23	Simulation under predict pressure loss coefficient by 4% on average
Vikas Kumar et al. (2003) Tube-in-Tube Helical Heat Exchanger	• CFD Code: PHOENICS • Turbulence Model: $k-\varepsilon$ • Mesh: 1 mm size, refined at the wall	For grid size: $2 \times 10^{-0.03}$ m, Nu: ±8.45% Error increases with an increase grid size.
Qiuwang Wang et al.(2008) Shell and Tube Type Heat exchanger	• CFD Code: FLUENT • Pressure-Velocity Coupling Scheme: SIMPLE • Meshing Software: Gambit	CFD overestimates overall heat transfer rate and pressure drop by 8.4% and 3.6% respectively than the estimated correlations.
Q.W. Dong et al. (2007) ROD Baffle Heat Exchanger	• CFD Code: FLUENT • Turbulence Model: $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLE, Second Order Upwind Scheme • Meshing software: GAMBIT • Mesh: Collocated grid	CFD underestimates heat transfer and fluid flow by 10% and pressure drop by 20%
Tri Lam Ngo et al. (2007) Micro Channel Heat Exchanger	• CFD Code: FLUENT	S-shaped fins: ±16.6%. For Zigzag fins: ±13.5%.
Yongqing Wang et al. (2007) Shell and tube heat exchanger	• CFD Code: FLUENT • Turbulence Model: standard $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLE • Discretization: Second order upwind scheme • Meshing Software: GAMBIT	Heat transfer coefficient: 10% Pressure drop: 20%
Dong Eok Kim et al. (2008) Printed Circuit Heat Exchanger	• CFD Code: FLUENT • Meshing Software: GAMBIT 6.3 • Turbulence Model: standard $k-\varepsilon$ • Mesh: non-uniform	Simulation under predicted results by 10%
Koorosh Mohammadi et al. (2009) Shell and Tube Type Heat Exchanger	• CFD Code: FLUENT 6.1 • Pressure-Velocity Coupling Scheme: SIMPLE • Turbulence Model: RNG $k-\varepsilon$ • Mesh: hexahedral 1,200,000 cell	–
Ying Chi Tasai et al. (2009) Chevron Plate Type Heat Exchanger	• CFD Software: FLUENT 6.3 • Solid Modelling: PRO/E • Meshing software: GAMBIT • Turbulence Model: $k-\varepsilon$ realizable • Mesh: Unstructured tetrahedron	In good agreement
Yong Qing Wang et al. (2009) Plate Fin Heat Exchanger	• CFD Software: FLUENT • Segregated solver approach • Pressure-Velocity Coupling Scheme: SIMPLE • Discretization :Second Order Upwind • Meshing Software: GAMBIT • Mesh: collocated	Simulation under predict results by 5%
Pedro F. Lisboa et al. (2010) Kenics Static Mixer as a Heat Exchanger	• CFD Code: FLUENT • Turbulence Models: RNG $k-\varepsilon$, $k-\varepsilon$, $k-\omega$	±10%
Xiao-Hong Han et al. (2010) Chevron Plate Heat Exchanger	• Turbulence Model: RNG $k-\varepsilon$ • Meshing Software: GAMBIT • Mesh: Non-structured, tetrahedral	±35%
Yu-Ling Shi et al. (2010) Sparse Tubular Heat Exchanger	• CFD code: FLUENT 6.3 • Turbulence Model: Standard $k-\varepsilon$ • Mesh: 2 (million) Grids	±12 %
Junjie Ji et al.(2011) Gas Fired Tubular Heat Exchanger	• CFD Code: FLUENT 6.3; user defined code coupling of heat transfer between air side and gas side. • Turbulence Model: SST $k-\varepsilon$ for reaction zone • Mesh: 3 million cells	Static pressure: 1%, 9.9% Total heat rate: 1.6% Air temperature rise: 2.1%

corroborated with the experimental results presented by Kays and London [46] and pressure drop was defined as follows:

$$\Delta p = 4f' \frac{LG^2}{d_e 2\rho} \tag{3}$$

Inputting the geometric parameters of the model yielded the pressure loss per unit length. Experimental and simulation results highlighted the accuracy associated with using a simplified model as presented on a CFD package to study the pressure losses in any heat exchanger.

Next year, Pedro F. Lisboa et al. [47] considered the heat transfer and fluid flow characteristics of the Kenics static mixer which is utilized in the process industries for the formation of supercritical carbon dioxide. A CFD simulation was done using FLUENT to analyze the impact of pressure values on the heat transfer rate. Three different turbulence field models $k-\varepsilon$, RNG $k-\varepsilon$, and $k-\varepsilon$ were utilized to investigate the thermodynamic behavior of the static mixer. An increasing trend was observed in the pressure drop for a rising fluid velocity with an aspect ratio set below 3. A comparison with ordinary pipe mixer concluded that the static mixer is 3 times more efficient than ordinary pipe type mixer. Further, it was revealed that results derived using RNG $k-\varepsilon$ model yield closest results to the experimental observations exhibiting a maximum deviation of 10% when using Blasius Equations.

Yu-Ling Shi et al. [48] employed the semi porous media approach to study sparse tubular heat exchanger. This approach reduces the computation cost and is simpler in nature than the user defined function porous media, the general porous media and the real tube models. A comparison between eight designs validated the viability of the adopted methodology for predicting pressures inside the exchanger. Similarly, in an effort to reduce the computational cost for industrial scale gas fired tubular heat exchangers, Wang et al. [3] and Gu et al. [49] proposed a simplification by considering the entire exchanger as a single duct unit. Carrying this forward, in 2011, Junjie Ji et al. [50] carried a full grid, hybrid simulation with 1 D + 2 D and 3D models. The tubes were divided into a reaction zone and a non reaction zone while gas side and the air side were analyzed using separate schemes. A comparison between the results established hybrid technique as a viable method to analyze and optimize large scale industrial exchangers, keeping the computational cost to a minimum.

3.4. Thermal analysis

The very purpose of a heat exchanger is to enhance the heat transfer between two fluids. This reduces the energy requirements and helps make a process more efficient both in terms of production and economics. With the help of CFD analysis, researchers have approached thermal characteristics of heat exchangers from two points of view i.e. thermal coefficients and effect of physical features on heat transfer rate as shown in Table 4a and b below.

3.4.1. Influence on thermal parameters

A basic strategy to evaluate an exchanger's performance is through the study of parameters like Nusselt Number, Dean's number, Prandtl number, Friction factor, Colburn factor etc and their impact on the overall heat transfer coefficient of the heat exchanger [2,7,51,52].

Vikas Kumar et al. [35] considered a cross flow air to air tube type heat exchanger and observed the relation between heat transfer and Dean's Number keeping the flow rate constant in the outer region. An increase in heat transfer from 800 to 900 W/m2 was observed with a rise in inner De from 3500 to 6000. For the outer De, an increase from 500 to 1000 yielded an increase from 600 to 950 W/m2. It was assumed that heat lost occurred through convection from the top surface of the exchanger only whereas all radiation losses were neglected in the study. Due to symmetry in geometrical construction, a section of heat exchanger was considered for CFD analysis.

Carla S. Fernandes et al. [51], 2005, presented different correlations for Nu, Re and Pr and discussed the fluid entry effect on average Nu for stirred yogurt during cooling in a PHE. Herschel–Bulkley model was employed and numerical analysis carried out by first considering and then ignoring the temperature effect on the viscosity. A comparative analysis with correlations present in past literature concluded that Nu experiences an increase with an increase in the Re. The study made it evident that for yoghurt simulation, CPU time can be greatly reduced when the influence of temperature is ignored. Similarly, Athanasios G. Kanaris et al. [2], focused on establishing the significance of CFD as a method to study the corrugated type PHE with stainless steel plates and a herringbone structure for single pass counter current flow. The temperature distribution was shown to remain constant in the *y*-direction i.e. along the width of the channel. The author established that Nu/Nuo increases with Re while friction factor decreases for trapezoidal corrugation, which is greater than sinusoidal

Table 4a
CFD simulations on heat exchangers by studying thermal coefficients.

Authors/Type	Methodology	Difference b/w Experimental and Simulation Results
Vikas Kumar et al.(2003) Cross flow Air to Air Tube Type Heat Exchanger	• CFD Code: PHOENICS	In good agreement.
Carla S. Fernandes et al. (2005) Plate Heat Exchanger	• CFD Code: POLYFLOW	Compared with correlation by Afonso et al. (2003): Considering the temperature effect on viscosity: 3.6% Disregarding the temperature effect on viscosity: 8.9%
A G. Kanaris et al. (2006) Corrugated Plate Heat Exchanger	• CFD Code: CFX 10.0 • Mesh: unstructured tetrahedral and prism elements • Turbulence Model: SST	Generally, simulation over predicts the values
J.S. Jayakumar et al. (2007) Helically Coiled heat exchangers	• CFD Code: FLUENT 6.2 • Turbulence Model: Realizable $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: SIMPLEC Meshing Software: GAMBIT 6.2 • Mesh: Structured (9.963e8 cells m^{-3})	5%
A.G. Kanaris et al. (2008) Plate Heat Exchanger with Undulated Surfaces	• CFD Code: ANSYS CFX 10.0 • Turbulence Model: RSM • Meshing Software: ANSYS ICEM CFD • Mesh: unstructured tetrahedral and prism elements	10%
Andrew M. Hayes et al. (2008) Matrix Heat Exchanger	• CFD Code: FLUENT • LTE (Local thermal equilibrium model), LTNE(Local thermal non equilibrium model) • Auxiliary Technique: Porous medium methodology	Results found to be in good agreement
In Hun Kim et al. (2009) Printed Circuit Heat Exchanger	• CFD Code: FLUENT • Auxiliary Technique: KAIST helium test loop system	No such comparison found
K. Kritikos et al. (2009) Staggered Elliptic Tube Heat Exchanger	• CFD Code: CFD 3D solver • Model: Porous medium methodology • Mesh: 45,584 cells	The experimental and simulation results did not show a discrepancy of more than 5%.
L. Sheikh Ismail et al. (2009) Plate Fin Heat Exchanger	• CFD Code: FLUENT • Turbulence Mode: Standard $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: Semi implicit SIMPLER • Mesh: 2,20,000 elements for best results	Rectangular Fin, Low Re: *j*: 2%; *f*: 9% Wavy Fins, $1000 < Re < 15000$ *j*: 10%; *f*: 20%
S. Freund et al. (2010) Plate Heat Exchanger	• CFD Code: ANSYS ,CFX 11 • Turbulence Models: SST and RSM • Meshing Software: ICEM • Mesh: Fine hexahedra	SST model: 33% RSM model: 25%

Table 4b
CFD simulations of heat exchangers in the field of design analysis through thermal study.

Authors/Type	Methodology	Difference b/w Experimental and Simulation Results
S. Knudsen et al. (2003) Vertical Mantle Heat Exchanger	• CFD Code: FLUENT 6.0 • Turbulence Model: $k-\varepsilon$ • Pressure-Velocity Coupling Scheme: PISO	A good agreement observed
Kwasi Foli et al. (2005) Micro Channel Heat Exchanger	• CFD Code: CFD-ACE+ • Pressure-Velocity Coupling Scheme: SIMPLEC	Close agreement observed
Vijaisri Nagarajan et al. (2008) Bayonet Type Heat Exchanger	• CFD Code: FLUENT 6.2.16 • Meshing Software: GAMBIT 2.2.30 • Mesh: Tetrahedral	Different geometrical configurations yielded different experimental results
A.-M. Gustafsson et al. (2009) Ground Water Filled- Bore Hole Type Heat Exchanger	• CFD Code: FLUENT • Mesh: 634,000 hexahedron and wedge shaped cells	±5%
Vikas Bansal et al. (2009) Earth-Pipe Air Heat Exchanger	• CFD Code: FLUENT 6.3	0%-9% for the different sections of the exchanger
Charbel Habchia et al. (2010) Novel Multifunctional Heat Exchanger	• CFD Code: FLUENT 6.3 • Turbulence Model: $k-\varepsilon$, RSM	A good agreement observed
Ender Ozden et al. (2010) Shell and Tube Type Heat Exchanger	• CFD Code: ANSYS FLUENT 6.3 • Turbulence Models: $k-\varepsilon$ (Standard and Realizable) • Mesh: Quadrilateral and tetragonal hybrid elements-coarse cells (70,000) and fine cells (1360,000) • Meshing software: GAMBIT	CFD overestimates analytical results by 36%.
Yupeng Wu et al. (2010) Slinky Ground Source Heat Exchanger	• CFD Code: FLUENT • Model: transient 3 dimensional sensible heat transfer model	A good agreement observed between the experimental and the simulation results
V. Antony et al. (2011) Shell and Tube Type Heat Exchanger	• CFD Code: FLUENT • Turbulence Model: $k-\varepsilon$ • Descretization: Second Order Upwind Scheme • Pressure-Velocity Coupling Scheme: PISO • Meshing software: GAMBIT	–

corrugation for same Re. A good agreement was found with the results presented by Vlasogiannis et al. [53].

Carrying their study forward, Kanaris et al. determined a general suitable mechanism of designing a PHE with undulated surfaces in 2008. The results were expressed in the form of Nu and friction factor for the optimization function. These were compared with the literature and optimal characteristics were proposed for an array of Re for two values of weighing factor. Through the CFD model, it was observed that the when the distance between channel plates reduces the positive performance of PHE increases. PHE performance can also be increased by decreasing the channel aspect ratio and increasing the angle of the attack [54].

Marko Lyytikäinen et al. [52] used depth-averaged flow and energy equations to perform CFD simulation. Equations were limited to 2D from 3D but friction factor and heat transfer through plates was accounted. A comparison was made between 2D simulation and their results with the 3D model of five unlike corrugation angles and lengths. The 2D simulation results presented a relatively good summary of pressure loss and temperature variation with respect to corrugation angle and length. 30 more different geometries were simulated using fast 2D approach and it was observed that temperature change due to geometry variation is not as significant as the pressure drop is.

In 2010 [14], S. Freund et al. developed a technique employing infrared thermography to investigate the heat transfer coefficients of a similar PHE. The surface mapping of the exchanger revealed presence of minimum heat transfer coefficient at the point of intersection of two meshes and lines of local maximum parallel to the length of corrugations. A CDF analysis was performed that demonstrated a better convergence mode for RSM-EASM turbulence model. The authors pointed out that despite the possible under estimation of heat transfer coefficient found via CFD, applying these results in heat exchanger design would only result in a better exchanger performance than predicted; thus, the significance and possible use of CFD in heat exchanger design remains non-detrimental and unquestionable. Xiao-Hong Han et al. [45] stated for a chevron PHE that highest temperature appears in the region of upper port while lowest temperature is found at the inflow of the cold fluid. An increase in temperature gradient is obtained downstream of the flow.

J.S. Jayakumar et al. [7], 2007 studied thermal behavior of fluid media in a helical pipe followed by a shell and tube heat exchanger with helically coiled tube bundles. The thermal properties were presented in the form of Nu and De, considering them to be dependent on temperature alone. Firstly, three analyses were carried out on the helical pipe; one for constant properties at ambient temperature, second for constant properties at mean temperature of the fluid and lastly for temperature dependent properties. For modeling the latter, equations were obtained via regression analysis in statistical tool of Matlab. The author concluded that values of constants (specific heat, thermal conductivity, density etc.) vary with temperature and assuming them constant over a larger temperature difference produces in accurate results. An increasing trend for the overall heat transfer coefficient was obtained for increasing De with temperature dependant properties yielding closest results to the experimental outcomes. A correlation was presented to give the inner heat transfer coefficient, valid for $2000 < \mathrm{De} < 12{,}000$, as stated below:

$$\mathrm{Nu} = 0.025\ \mathrm{De}^{0.9112}\mathrm{Pr}^{0.4} \tag{4}$$

In 2008, Andrew M. Hayes et al. [55] modeled a matrix heat exchanger using porous medium for heat transfer. Venkatarathnam and Sarangi [56] had researched on the heat transfer and fluid flow characteristics in actual micro heat exchangers. The authors, utilizing the results of that study stated a correlation between Re and the Nu number for the matrix heat exchanger implying porous media given as under:

$$\mathrm{Nu} = 0.397\mathrm{Re}^{0.652} \tag{5}$$

To evaluate the uncertainties in the coefficient of heat transfer, the Kline–McClintock method of uncertainty calculation was used which was found to be 2.7%.

Gongnan Xie et al. [57], 2009, evaluated the Nu and friction factor for three types of fin and tube heat exchangers i.e. with slit fins, longitudinal fins and plain fins. Study was carried on to observe the difference between laminar and turbulent heat transfer for larger diameter of tubes and comparatively larger number of rows of tubes.

In Hun Kim et al. [58] employed the KAIST Helium test loop system and the CFD techniques to calculate the thermal parameters at the inlet and outlet of a Printed Circuit Board Heat Exchanger. The intrinsic properties of PCHE were analyzed previously by Gezelius et al. [59] in 2004 using CO2 as a working fluid. Hun Kim et al. presented correlations for global fanning factor, global Nu and Local Nu were proposed and it was found that the local Nu found through the CFD simulations is best to carry out the analysis:

$$\mathrm{Nu}_{-p} = 4.065 + 0.00305 \times \mathrm{Re}_{-p} \quad 350 < \mathrm{Re}_{-p} < 800 \quad \mathrm{Pr}_{-p} = 0.66 \tag{6}$$

K. Kritikos et al. [4], 2009 adopted the porous medium and full scale geometry techniques to analyze the heat transfer in elliptical tube staggered heat exchanger. The results were quantified in form of graphs of Nu with fluid velocity and the tube number. An increase in the $\mathrm{Nu}/\mathrm{Nu}_{ref}$ by a factor of 0.2 was observed with an increase in velocity whereas it was concluded that the first two tubes played a primary role in the thermal characteristics of the exchanger.

A study of the f and j was undertaken by L. Sheikh Ismail et al. [60] for an offset and wavy fins compact plate fin heat exchanger. f and j were seen to increase with increasing height to fin ratio while for the same height to fin ratio, f and j were found to be higher for smaller Re. Owing to under estimation of results, it was concluded that all results should to be multiplied by 1.6.

3.4.2. Design optimization

Ever since its advent, CFD has proven to be an effective tool in the design optimization of the heat exchangers by studying thermal properties. It has been employed to study different modifications [10,61], compare results [9,62] and present the best possible combination of variables to ensure optimum performance [18,63].

In 2003, S. Knudsen et al. [13] discussed the effect of inlet positions of the mantle on the thermal behavior for a vertical mantle heat exchanger used in solar domestic hot water systems. By assuming the symmetry along the mid plane, only half of the model was considered for CFD simulation to reduce the computational time. It was concluded that it is more efficient to have the inlet located at the top for a high inlet-temperature to the mantle while for low temperature inlet, moving the inlet down yields better performance. In another study following year, the authors extended the same work and concluded that the heat transfer rate for the exchanger is dependent more on inlet temperature of the mantle as compared to the temperature of the core tank [18].

Kwasi Foli et al. [63] stated that nature of the fluid, operating conditions and geometry play a major role in defining the performance of a micro channel heat exchanger. CFD analysis was coupled with the analytical and multi-objective genetic algorithms methods respectively to determine the optimum design and the results obtained from the two approaches were compared. The latter was found more useful as it did not only help find the optimum dimensions but the shape of the optimum design structure too. The performance of the exchanger was found greatly influenced by the aspect ratio, defined as ratio of height to the width of a channel. In case of a variable volume of micro channels, for an increase in the aspect ratio from 10 to 100, the pressure drop and the heat flux experienced a steep drop initially attaining a stable value at 100. However, the heat transfer rate increased slightly and stabilized at an overall aspect ratio of 30. On the contrary, for a constant volume micro channel, pressure loss and the heat transfer tends to increase with increasing aspect ratio whereas; the heat flux experiences a drop. In the same year, Malcom J Andrews et al. [61] performed CFD simulation to study the characteristics of a helix heat exchanger. Baffle geometry and position of the nozzle for inlet and exit configurations were primarily considered together with three different helix angles measured relative to radial axis. It was established that the much desired plug flow is observed at the outer region of the exchanger which increases with the helix angle. A comparison between the results of the pressure drop obtained through ABB Lummus Heat Transfer, Bloomfield, New Jersey correlations and simulation shows a good agreement; however more effort is required to estimate the impact of nozzle losses on helixchanger performance.

In 2008, S. Freund et al. [14] developed a contact free and fluid autonomous method for measuring convective coefficient of heat transfer of a PHE. The method relied on experimental data of the outside body temperature subjected to periodic heating, measured through infrared camera, and on the heat exchanger wall. The pattern of the heat transfer coefficient was found to be very distinct where heat transfer was least i.e. at the intersecting points of the corrugated body and lines of maximum value adjacent to the corrugation. The whole plate was analyzed (PHE) to measure the macro scale flow pattern and no maldistribution was accounted.

Vijaisri Nagarajan et al. [62] evaluated the impact of exchanger design on decomposition behavior of sulphur trioxide in bayonet type heat exchanger. They concluded that low decomposition is achieved at temperatures below 1000 K, and sufficient above 1073 K. Considering these results, to predict a better design, thermal modeling was undertaken to evaluate the decomposition process. The impact of cylindrical, spherical, cubical and hollow pellets arranged inside the packed bed was studied on the overall exchanger performance. Pellets were composed of silicon with platinum catalyst while the decomposer was made of silicon carbide and quartz. A comparative analysis of the decomposition percentage of SO_3 for the packed bed and the porous media approach was executed. Results yielded that spherical pellets provide the maximum decomposition percentage, in this case, 61% for 5 ml/min.

For shell and tube heat exchanger, Qiuwang Wang et al. [37] performed the comparison of combined multiple shell-pass shell and tube type heat exchanger with segmented baffles using CFD simulation. The comparative analysis of the results obtained from simulation of the two distinct models revealed that under similar mass and heat transfer rate with same pressure drop in the shell side the overall heat transfer rate of CMSP-STHX is 5.6% higher than the SG-STHX. Ender Ozden et al. [6] performed the CFD simulation in a shell and tube heat exchanger to estimate the effect of baffle spacing, cut and shell diameter on the heat transfer and pressure drop characteristics. To evaluate the performance, different baffles and turbulence models were examined. In order to determine the best turbulence model a comparison of Bell Delaware method and Kern method [64] were carried out and two important conclusions were deduced: Kern method always underestimates the heat transfer coefficient while Bell Delaware [65] method underestimates the heat transfer rate of varying spaced baffles, showing good agreement with the heat transfer characteristics of equally spaced baffles. It was further established that a baffle cut of 25% gives superior results for heat transfer. V. Antony et al. [66] considered a shell and tube type heat exchanger based on the PCM Module for the cooling purposes. A modular type arrangement was used for the exchanger with several modules arranged over one another consisting of air spacers between two successive modules. Transient and steady state CFD modeling was carried for a single module and two consecutive air spacers found very effective to study the pressure drop parameters around the modules and the temperature variations of the working fluid. The results obtained were important to evaluate the solidification characteristics of the PCM and to prove the suitability of the adopted geometrical dimensions. It was concluded that air spacers improve the heat transfer rate significantly, decreasing the solidification time by 50–75%, with a more pronounced effect at lower air velocities.

A borehole heat exchanger using ground water as working media was studied by A.M. Gustafsson et al., 2009 [8]. The objective was to estimate the result of convection in the overall heat transfer. A CFD simulation of the proposed U shaped model was carried out based on the study of Desai and Littlefield [67] to evaluate the heat transfer rate when induced convection was considered along with the conduction heat transfer. A constant temperature gradient was developed by imposing a higher constant temperature on the U shaped wall and a lower temperature on the exterior bedrock wall, which induced a velocity flow in the ground water. This resulted in an increased heat transfer as compared to still water conditions. Further, effect of gradient in the heat transfer and mean temperature level was established. To evaluate the thermal resistance, convective heat flow, actual heat transfer rate and borehole temperature must be taken into account.

Li-Zhi Zhang et al. [12] investigated a cross flow air to air heat exchanger with plate-fin cores for thermal deterioration over time. For a flow rate between 100 and 200 m^3/h and a Re of 200 the change in various physical parameters were observed. There was a variation of ± 0.1 °C in temperature and $\pm 1\%$ in flow rate. A deviation in the overall effectiveness of the exchanger was obtained within $\pm 4.5\%$ while hardly any change in the fresh inlet and outlet temperatures was encountered; it was less than 0.1%.

Vikas Bansal et al. [68] performed a study on Earth Pipe Air Heat exchanger (EPAHE) to understand the effect of the pipe material and the velocity of air on the heat transfer through the exchanger. Earlier, Kumar et al. [69] had investigated the conservation potential of an earth air pipe exchanger installed for a building without an air conditioning system. Ghosal et al. [70] had presented a model to evaluate the performance of earth to air heat exchanger (EAHE) coupled with a green house. Additionally Ajmi et al. [71] conducted his research on the cooling ability of earth to air heat used in the desert areas. Utilizing the results of all these studies, Kumar conducted a CFD simulation. The velocity of air was observed to be the primary parameter of optimizing the heat transfer. An improvement in the cooling from 12.7 °C to 8 °C was achieved for an increase in air velocity from 2 m/s to 5 m/s. The study concluded that a cheaper material may be employed for the pipe as the effect of using steel or PVC on the effectiveness was found to be almost negligible.

In 2010, Charbel Habchi et al. [9] observed heat transfer parameters and the turbulent mixing in the multifunctional heat exchanger under three different geometrical configurations. The Laser Doppler analysis and the CFD approach were used to determine the consequences of the vorticity produced by the trapezoidal tabs installed in three different inclinations. The first arrangement contained a circular pipe having seven trapezoidal tabs, the second arrangement was of reversed tab array, and the third configuration included the tabs being rotated at 45°. A turbulent flow field was considered for the analysis using Re values of 7500, 10000, 12500 and 150000 respectively. The reversed configuration proved to be the most efficient one for micro mixing and meso mixing having the efficiency values of 50% and the 25% respectively. However it also increased the power consumption by 40%.

Yupeng Wu et al. [72], on the other hand, analyzed the horizontal-coupled slinky ground source heat exchanger in the UK climate. Florides and Kalogirou [73] evaluated the performance characteristics of a ground coupled heat exchanger. They concluded that atmospheric conditions have a significant impact on the exchanger's performance below and its influence should be considered while designing Ground source heat exchangers. Considering the observations of this research, Yupeng Wu carried out a CFD simulation to study thermal performance of straight and slinky heat exchanger focusing the effect of coil diameter and coil central interval difference. The following transient three dimensional sensible heat transfer model (FLUENT) was employed:

$$\frac{\partial(\rho T)}{\rho t} + \nabla \cdot \left(\rho \vec{V}\, T - \frac{k}{c} \nabla T \right) \;=\; \cdot \frac{q}{c} \tag{7}$$

For comparison, an experimental setup including four horizontal-coupled slinky heat exchanger loops was utilized. After an observation of two months, it was found that COP reduces from 2.7 to 2.5 i.e. heat extraction of both, straight and slinky heat exchangers decreases over time. Amongst the two, specific heat extraction of straight heat exchanger is greater than the slinky heat exchanger, whereas, the latter's ability of heat extraction per unit length is higher. The study concluded that specific heat extraction is independent of coil diameter and an inverse relation exists between the coil interval distance and heat extraction per unit length.

4. Conclusions

Conventional methods used for the design and development of Heat Exchangers are largely tedious and expensive in today's competitive market. CFD has emerged as a cost effective alternative and it provides speedy solution to heat exchanger design and optimization. Easily accessible general purpose CFD commercial softwares can fulfill the requirements of CFD analysis of various types of heat exchangers including but not limited to Plate, Shell and Tube, Vertical Mantle, Compact and Printed Circuit Board Exchangers. These are flexible enough to accommodate any kind of analysis requirement ranging from prediction of fluid flow behavior to complete heat exchanger design and optimization involving a wide range of turbulence models and integrating schemes available in CFD softwares. Of these, the $k - \varepsilon$ turbulence model has been most widely employed for heat exchanger design optimization. The simulations generally yield results within good agreement with the experimental studies ranging from 2% to 10% while in some exceptional cases, vary up to 36%. In case of large deviations, user defined sub routines specific to the design problem may become necessary. Dependability and reliability of CFD results has reached a point making it an integral part of all design processes, leading towards eliminating the need of prototyping.

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